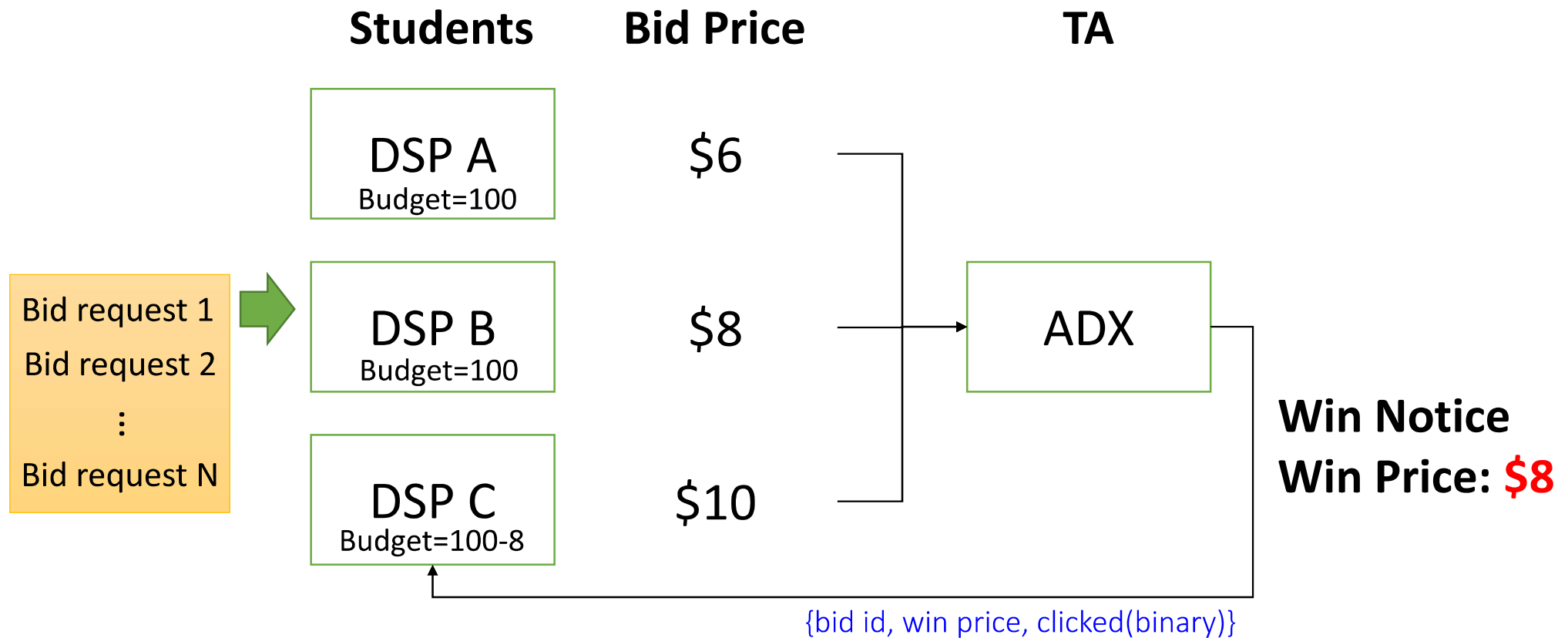


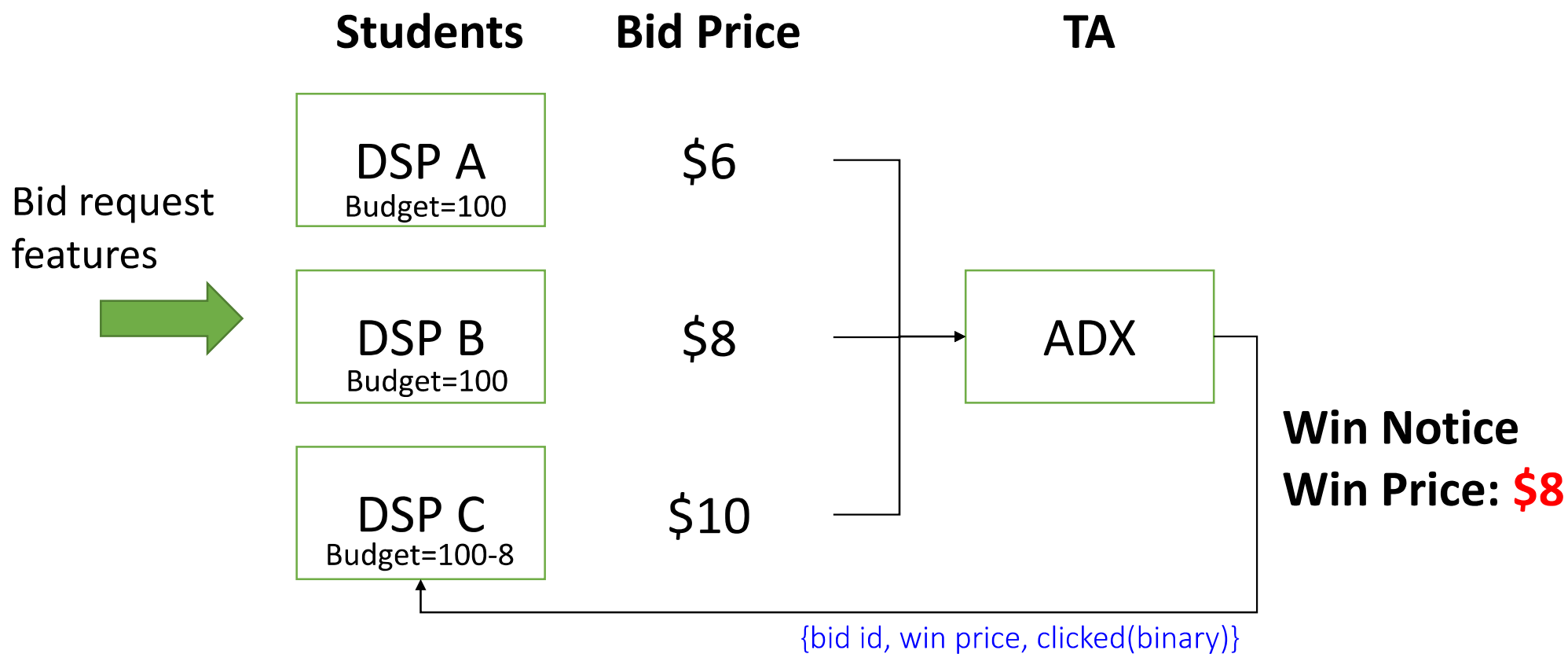
Project Choice 2 – Individual Track

- Join an advertisement bidding game individually
 - You are a bidding agent for an advertiser
 - Your goal is to maximize KPI by winning bids with a given budget
- **Report: 3-4 pages**
 - ACM template
 - Double column
 - Figures, tables, equations, etc
- Participants are prohibited to share information with each other

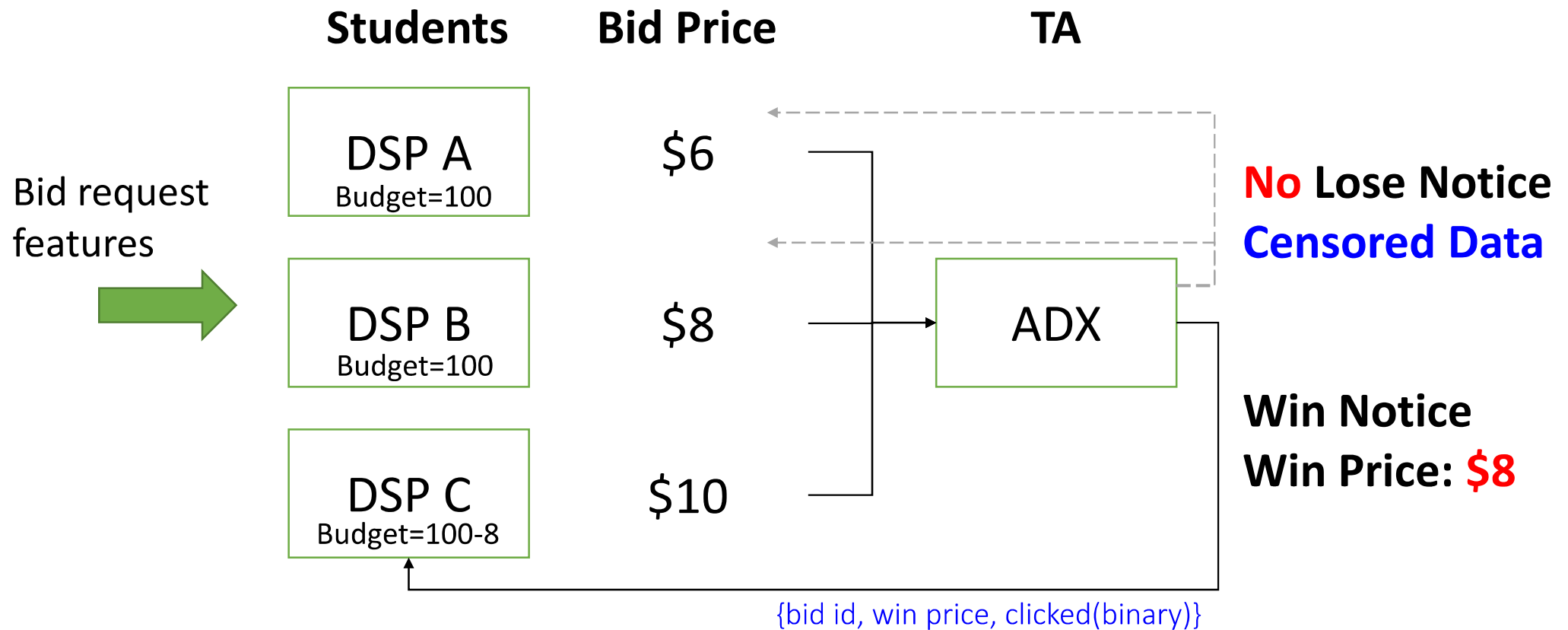
Bid Request Stream



Second-Price Auction (SPA)



Censored Data



Bidding Strategy

- **Given** a stream of bid requests and their features, ad features (**budget**, ad content, target audience), etc
- **Determine** bid prices for each bid request
- **Such that**

$$\begin{aligned} & \max \text{KPI} \\ \text{s.t. } & \sum \text{cost} \leq \text{Budget} \end{aligned}$$





KPI

- Win = display an ad to audience (impression+1) -> if clicked (click+1)
- For this project: **#clicks > #impressions**
- The ranking results among participants will reflect on final scores

Dataset and Procedure

- 4-day training and 1st, 2nd, and 3rd testing sets
 - Consecutive days, millions of bid requests
- All submission deadline at 23:59 Taipei time

**No result exchange
between students!**

Date	Student Deadline	TA Release
6/4	Registration	training and 1 st testing
6/11	1 st day bidding price	1 st day result and 2 nd day testing
6/18	2 nd day bidding price	2 nd day result and 3 rd day testing
6/25	3 rd day bidding price and report	Final results

Basic Statistics

	Training	Testing (total)
#Impressions	1,760,309	~130M
#Clicks	1,319	~1,000
Average CTR	0.07%	-
Average CPM	68.88	-

See `feature_meaning.xlsx` for feature details.

Bidding Submission Data Format

- Row data
 - Testing: {bid id, feature}
 - Submit: {bid id, bid price}
 - Feedback:
 - If you win a bid: {bid id, win price, clicked(binary)}
 - Receive nothing if lose
- File name: {student-id}_day{number}.csv
 - E.g., 0586004_day1.csv for the 1st testing day
- Any data read fail on our side (e.g., naming or formatting) will cause you 0 grade

No result exchange
between students!

Other Game Rules

- All prices in integer
- Daily budget: \$5,000 (i.e., \$5,000 for each testing date individually)
- Winner is charged when winning a bid request
- For tie-breaking
 - Randomly choose a winner
 - Win price:
 - **If there is a distinct second-high bid price:** the distinct second-high bid price
 - **Else if everyone ties and the highest bid price > 2 :** the highest bid price – \$1
 - **Else:** no winner

Main Components

- Same as team project if not specified
- Dataset (20%)
- Model (20%)
 - How you utilize online updating?
- Literature (20%)
- Bidding Results
 - (report) Training results on the first 4 days and testing results (20%)
 - Performance test, ablation tests, hyperparameter settings
 - Ranking results among all participants for the 1st to the 3rd testing days (20%)
 - Click (15%) and impression (5%)

General Rules for Using GAI

- Plagiarism and the Use of ChatGPT or similar LLMs Papers that include text generated from a large-scale language model (LLM) such as ChatGPT are prohibited unless the produced text is presented as a part of the paper's experimental analysis. Note that this policy does not prohibit authors from using LLMs for editing or polishing author-written text. [AAAI' 25]
- Generative AI models, including ChatGPT, BARD, LLaMA, or similar LLMs, do not satisfy the criteria for authorship of papers published in IJCAI 2025. If authors use an LLM in any part of the paper-writing process they assume full responsibility for all content, including checking for plagiarism and correctness of the entire submission. [IJCAI' 25]
- Justify your contributions against GAI if you are collaborating with them



Make Your Choice!

- <https://reurl.cc/dQolb2>
- Registration closes on 6/4



	A	B	C	D
1	Team ID	Student ID	Student ID	Student ID
2				
3				
4				
5				
6				
7				
8				
9				
10				

+

≡

Team ▾

Individual ▾

	A	B	C
1	Student ID		
2			
3			
4			
5			
6			
7			
8			
9			
10			

+

≡

Team ▾

Individual ▾